When Einstein discovered the wave/particle duality of light, he did not realize that an electron or proton might also behave sometimes like a particle and sometimes like a wave. Nobody recognized it until the mid-1920s when Louis de Broglie raised it as a conjecture and then Erwin Schrödinger used it as a foundation for a full set of quantum mechanical laws, laws in which an electron is a wave of probability. Probability for what? For the location of a particle. These "new" quantum mechanical laws (which have been enormously successful in explaining how T electrons, protons, atoms, and molecules behave) will not concern us much in this book. However, from time to time a few of their features will be important. In this chapter, the important feature is electron degeneracy.

When Eddington wrote his book, electron degeneracy had not yet been predicted, and it was not possible to compute correctly how rock or other materials will respond if compressed to the ultra-high densities of Sirius B. With the laws of electron degeneracy in hand such computations were now possible, and they had been conceived and carried out by R. H. Fowler in his 1926 article.

According to Fowler's computations, because the electrons in Sirius B and other white-dwarf stars have been compressed into such tiny cells, their degeneracy pressure is far larger than their thermal (heat-induced) pressure. Accordingly, when Sirius B cools off, its minuscule thermal pressure will disappear, but its enormous degeneracy pressure will remain and will continue to support it against gravity.

Thus, the resolution of Eddington's white-dwarf paradox was twofold: (1) Sirius B is not supported against its own gravity primarily by thermal pressure as everyone had thought before the advent of the new quantum mechanics; rather, it is supported primarily by degeneracy pressure. (2) When Sirius B cools off, it need not reexpand to the density of rock in order to support itself; rather, it will continue to be supported quite satisfactorily by degeneracy pressure at its present density of 4 million grams per cubic centimeter.

Chandrasekhar, reading these things and studying their mathematical formulations in the library in Madras, was enchanted. This was his first contact with modern astronomy, and he was finding here, side by side, deep